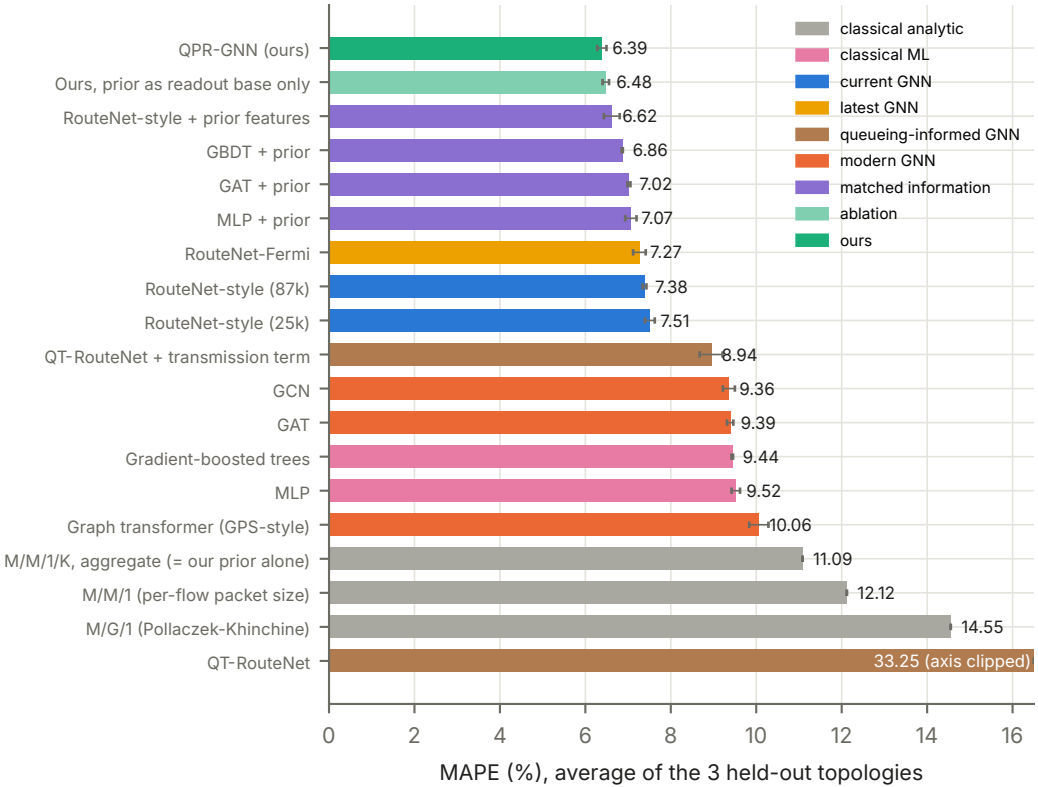
QPR-GNN (ours)
Ours, prior as readout base only
RouteNet-style + prior features
GBDT + prior
GAT + prior
MLP + prior
RouteNet-Fermi
RouteNet-style (87k)
RouteNet-style (25k)
QT-RouteNet + transmission term
GCN
GAT
Gradient-boosted trees
MLP
Graph transformer (GPS-style)
M/M/1/K, aggregate (= our prior alone)
M/M/1 (per-flow packet size)
M/G/1 (Pollaczek-Khinchine)
QT-RouteNet
6.39
6.48
6.62
6.86
7.02
7.07
7.27
7.38
7.51
8.94
9.36
9.39
9.44
9.52
10.06
11.09
12.12
14.55
33.25 (axis clipped)
classical analytic
classical ML
current GNN
latest GNN
queueing-informed GNN
modern GNN
matched information
ablation
ours
0
2
4
6
8
10
12
14
16
MAPE (%), average of the 3 held-out topologies